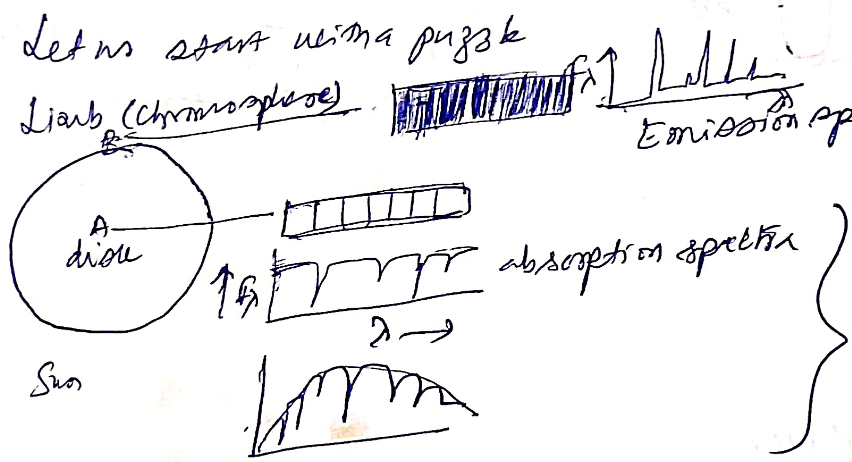
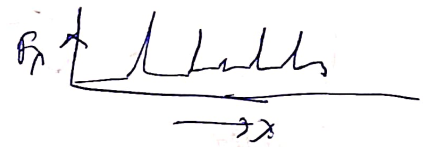
① Principles of Radiation transfer: Absorption and Emission of radiation.

21-22-23 March 2024

Let us start with a puzzle





Why is it that the spectra on the disk shows absorption and ~~disk~~ (Chromosphere) Limb (or disk edge) shows emission spectra.

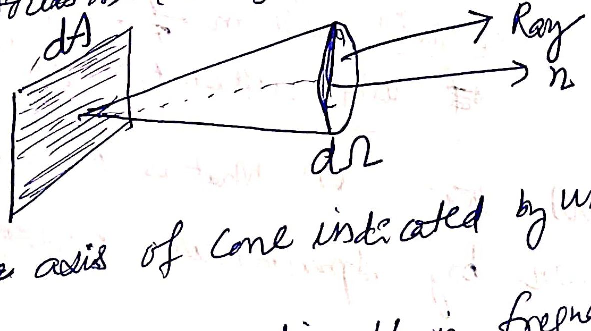
These things can be understood with the help of basic equations of radiative transfer

Let us introduce the basic notation of "Specific intensity" ~~of~~ or "brightness"

[~~focus bit~~ pay attention on units and dimension of various parameters]

Let us consider an area element dA, the radiation is entering this dA element from the left and going to the right at a time radiation going into solid angle $d\Omega$

Let us focus on the ~~light~~ radiation



with an ~~angle~~ axis of cone indicated by unit vector 'n' $\perp$ to the surface area dA

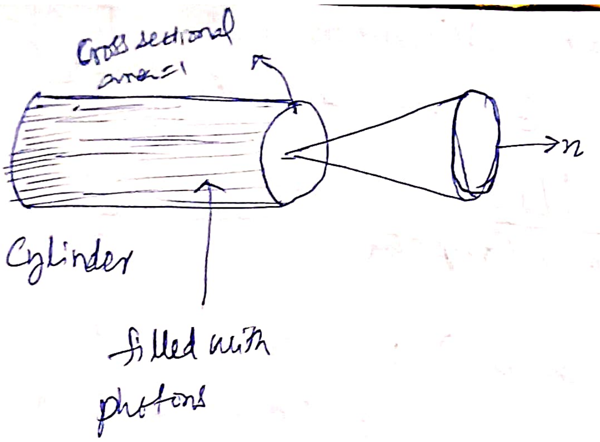
Energy crossing dA in time dt in frequency range $d\nu$, and entering into the solid angle $d\Omega$:

$$\boxed{dE = I_\nu \, dA \, dt \, d\nu \, d\Omega}$$

this energy defines the specific intensity or brightness I_ν

this brightness I_ν remains constant along the ray, and it won't decrease with distance (unlike the intensity)

②



$I_\nu = c\,U_\nu(\Omega)$

energy density per unit solid angle per freq interval.

Cylinder length = $c\,dt$

all the photons in elemental cylinder in time interval dt pass through the elemental surface and enters into the cone.

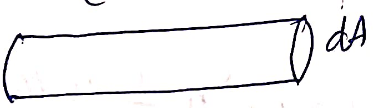
Emission coefficient

Spontaneous emission coefficient j_ν is the energy emitted per unit time per unit solid angle per unit volume

$$dE_\nu = j_\nu \, d\nu \, dV \, dt \, d\Omega$$

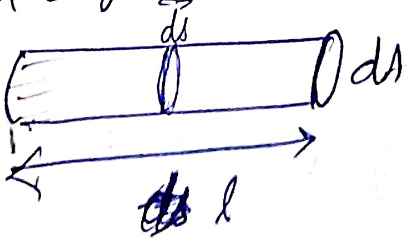
$$j_\nu = erg / cm^3 / s / Hz / sr^{-1}$$

Let us consider a cylinder with unit area dA, ~~we ask~~: What is the contribution of this medium (medium is a) to a beam of radiation ~~of~~ with cross section dA?



(or) What is the intensity added

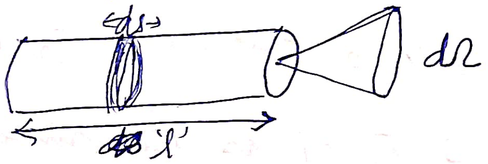
For ~~a beam of crossection dA, the~~ to the beam of radiation by spontaneous emission in a length ds is



$$dI_\nu = j_\nu \, ds$$

If radiation travels 2ds length then the emission contribution of that medium also increases by 2 times to specific intensity.

The loss of brightness in a beam as it travels a distance ds is



$dI_\nu = -\alpha_\nu I_\nu \, ds$; $-ve$ symbol denotes the loss of light.

Change in the specific intensity due to absorption | this much fraction is eaten up (taken by or absorbed by) that medium over a length ds.

- Since we have taken $-ve$, the value of α_ν is considered to be positive by convention

The dimension of $\alpha_\nu = cm^{-1}$

Let there exist 'n' number of absorbers per unit volume, and let the cross section of ~~the~~ each absorber be σ_ν (cm^2).

absorption coefficient: $\boxed{\alpha_\nu = n \sigma_\nu}$ $(cm^{-3} \cdot cm^2 = cm^{-1})$

Basic equation of Radiative transfer

$$\frac{dI_\nu}{ds} = -\alpha_\nu I_\nu + j_\nu$$

$\because dI_\nu = j_\nu ds + (-\alpha_\nu I_\nu ds)$

The optical depth τ_ν:

Integral of absorption coefficient over the length of the medium through which the radiation has passed ~~the~~ which is the cylinder with length 'L'.

for the length ds, the fraction that is absorbed is $\alpha_\nu I_\nu ds$, therefore the total amount absorbed over 'L' is

$$\tau_\nu(s) = \int_{s_0}^{s} \alpha_\nu(s') ds'$$

($cm^{-1} \cdot cm$ = no unit) dimensionless

(4) Optically thick medium $\tau_\nu > 1$

Optically thin medium $\tau_\nu < 1$

Brick wall is optically thick as it is opaque, τ_ν is ∞

air is optically thin and $\tau_\nu < 1$ as the air is transparent to the ~~...~~ radiation.

$$\boxed{\frac{dI_\nu}{ds} = \underbrace{-\alpha_\nu I_\nu}_{\text{absorption by the medium}} + \underbrace{j_\nu}_{\text{emission by the medium}}}$$

$$\frac{1}{\alpha_\nu}\frac{dI_\nu}{ds} = \frac{-\alpha_\nu I_\nu}{\alpha_\nu} + \frac{j_\nu}{\alpha_\nu}$$

Since $\alpha_\nu\, ds = d\tau_\nu$

$$\frac{j_\nu}{\alpha_\nu} = \frac{\text{emission coefficient}}{\text{absorption coeff.}} = \text{Source function } (S) = \frac{j_\nu}{\alpha_\nu}$$

$$\Rightarrow \boxed{\frac{dI_\nu}{d\tau_\nu} = -I_\nu + S_\nu}$$

we can introduce specific intensity associated with the body

Source function $S_\nu = \frac{j_\nu}{\alpha_\nu} = I_\nu^{Body} = I_\nu^B$

$$\frac{dI_\nu}{d\tau_\nu} = -I_\nu + I_\nu^B$$

$$\boxed{\frac{dI_\nu}{d\tau_\nu} = -(I_\nu - I_\nu^B)}$$

this is the contribution of the medium to the actual specific intensity passing through the cylinder

$$\frac{dI_\nu}{d\tau_\nu} = -(I_\nu - I_\nu^B) \longrightarrow ①$$

then $(I_\nu - I_\nu^B) = (I_\nu(0) - I_\nu^B)\, e^{-\tau_\nu}$

$\cancel{I_\nu = I_\nu(0)\, e^{-\tau_\nu} + I_\nu^B [1 - e^{-\tau_\nu}]}$

$$I_\nu = I_\nu^B + I_\nu(0) e^{-\tau_\nu} - I_\nu^B e^{-\tau_\nu}$$

$$\boxed{I_\nu = I_\nu(0) e^{-\tau_\nu} + I_\nu^B [1 - e^{-\tau_\nu}]}$$

I_ν = Specific intensity coming out of the medium [emergent intensity]

$I_\nu(0)$ = " " incident on the medium where $\tau_\nu = 0$ | $I_\nu(0)$ at $\tau_\nu = 0$

I_ν^B = Specific intensity added by the medium

$e^{-\tau_\nu}$ = attenuation factor

$I_\nu(0) e^{-\tau_\nu}$ = fraction of radiation survives (after exponential absorption) and survives without being absorbed

(or) = boundary radiation that passes through the medium and exponentially reduced as τ_ν increases

Summary

{ only local thermal emission survives

deeper inside the medium boundary effect vanishes but only local thermal emission survives

$\Rightarrow I_\nu = I_\nu^B + c e^{-\tau_\nu} \Rightarrow I_\nu \to I_\nu^B$ as $\tau_\nu \to \infty$

(b) $x = c e^{-\tau_\nu}$

$(I_\nu - I_\nu^B) = c e^{-\tau_\nu}$

$I_\nu = I_\nu^B + c e^{-\tau_\nu}$

If $\tau_\nu \to \infty \Rightarrow e^{-\tau_\nu} = e^{-\infty} = 0$

$I_\nu = I_\nu^B$

but in the deeper inside the medium ($\tau_\nu \to \infty$), boundary effect vanishes but only the thermal emission survives

$\Rightarrow$ local thermal emission survives

condition radiation

$c e^{-\tau_\nu} \Rightarrow$ has the memory of boundary condition; it decays exponentially

(a) as a boundary condition to determine C

$\tau_\nu = 0$ is chosen at the surface of the cloud or entry point of the radiation

let $x = I_\nu - I_\nu^B \rightarrow I$

I_ν^B is a constant at a fixed T

$$\frac{dx}{d\tau_\nu} = \frac{dI_\nu}{d\tau_\nu} + 0 \longrightarrow ②$$

$I_\nu^B = B_\nu(T)$

If T is constant $\frac{dI_\nu^B}{d\tau_\nu} = 0$

Substitute ② in ①

$$\frac{dx}{d\tau_\nu} = -(I_\nu - I_\nu^B)$$

$$\frac{dx}{d\tau_\nu} = -x$$

rearrange

$$\frac{dx}{x} = -d\tau_\nu$$

$$\int \frac{dx}{x} = -\int d\tau_\nu$$

$$\ln x = -\tau_\nu + c'$$

$$e^{\ln x} = e^{-\tau_\nu + c'}$$

$$x = e^{-\tau_\nu} \cdot e^{c'}$$

$$x = e^{-\tau_\nu} \cdot c$$

$$x = c \cdot e^{-\tau_\nu}$$

$$x = c \cdot e^{-\tau_\nu}$$

$$\boxed{x = c e^{-\tau_\nu}} \; ; \; c = \text{constant}$$

from the initial assumption

$$x = I_\nu - I_\nu^B$$

$$(I_\nu - I_\nu^B) = c e^{-\tau_\nu}$$

Apply boundary condition

At $\tau_\nu = 0$; $I_\nu(0) - I_\nu^B = C$

So $\boxed{(I_\nu - I_\nu^B) = (I_\nu(0) - I_\nu^B)\, e^{-\tau_\nu}}$

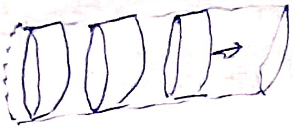
Since I_ν is a function of τ_ν.

$I_\nu = I_\nu(\tau_\nu)$: intensity depends on the optical depth $\Rightarrow$ similar to $f(x) = f(0)$ at $x = 0$

⑨

$I_\nu = I_\nu^B (1 - e^{-\tau_\nu})$

I_ν ~~$= I_\nu^B (1 - e^{-\tau_\nu})$~~



The medium add to the radiation.
But part of it is also absorbed by itself

When the medium is opaque $\tau_\nu >>1$ → Blackbody spectrum

1806 Kirchoff : LTE body emits at all λ's or (ν's)

spectrum is given by $I_\nu = I_\nu^B = \frac{2h\nu^3}{c^2} \left[\frac{1}{e^{h\nu/kT} - 1} \right]$ } given by Max Planck in 1900

universal BB function of T

Whereas I_ν ~~$= c U_\nu (\Omega)$~~

$I_\nu = c\, U_\nu (\Omega)$; U_ν is the energy density / unit solid angle / unit freq interval

$I_\nu = c\, U_\nu(\Omega)$

$U(\nu)\, d\nu = \frac{8\pi V h}{c^3} \cdot \frac{\nu^3}{(e^{h\nu/kT} - 1)} d\nu$ ← V should not be there

V = Volume of the cavity

$dE = I_\nu\, dA\, dt\, d\nu\, d\Omega$

I_ν = Energy/Area/time/freq/Solid angle

$U_\nu(\Omega) = I_\nu / c$ ✓

= Energy/Area/t/ν/Ω/m/s

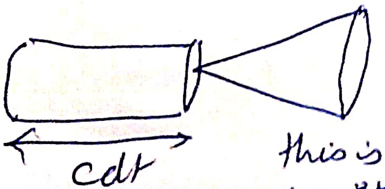
$U_\nu(\Omega)$ = (Energy/Volume)/ν/Ω

= Energy density/freq/Solid angle

$I_\nu = I_\nu^B = \frac{2h\nu^3}{c^2} \frac{1}{[e^{h\nu/kT} - 1]}$

actually 8π should be there in the numerator and c^3 in the denominator

this is because Planck's law is seen ~~as~~ always in ~~the~~ terms of energy density rather than Specific intensity



$I_\nu = c\, U_\nu(\Omega) \rightarrow \frac{I_\nu}{c} = U_\nu(\Omega)$

this is the energy density going into a particular unit solid angle $d\Omega$. per unit frequency

To find the total energy density, we have to integrate
$\frac{I_\nu}{c} = U_\nu(\Omega)$ over ~~all the~~ entire spherical volume
surrounded by the cavity → through this we pick up the
factor of 4π.

$$U(\nu)\,d\nu = \frac{I_\nu}{c}\,d\nu$$

$$= \frac{2h\nu^3}{c^2}\frac{1}{\left[\exp\left(\frac{h\nu}{KT}\right)-1\right]}\cdot 4\pi\,d\nu$$

$$= \frac{4\pi}{c}\cdot\frac{2h\nu^3}{c^2}\frac{1}{\left[\exp\left(\frac{h\nu}{KT}\right)-1\right]}$$

$$\boxed{U_\nu(\nu)\,d\nu = \frac{8\pi h\nu^3}{c^3}\frac{1}{\left[\exp\left(\frac{h\nu}{KT}\right)-1\right]}}$$

Planck's law in the energy density form

$\cancel{U_\nu = \frac{1}{c}I_\nu}$

$$dU_\nu = \frac{I_\nu}{c}\,d\Omega$$

or

$$dU_\nu(\Omega) = \frac{I_\nu}{c}d\Omega$$

I_ν coming in $d\Omega$ direction causes dU_ν energy density

$$\int dU_\nu = \int \frac{I_\nu}{c}\,d\Omega$$

$$U_\nu = \int \frac{I_\nu}{c}\,d\Omega$$

for the total energy density we just have summer all directions

If we assume isotropic radiation, I_ν = constant in all the directions

$$U_\nu = \frac{I_\nu}{c}\int d\Omega$$

$$U_\nu = \frac{I_\nu}{c}[4\pi]$$

In order to examine the Planck's law in the low vs higher wavelength or low or high frequency there is no meaning. Rather we do compare based on the low vs high energy.
here we have $h\nu$, but it can't be compared with other $h\nu$
but rather it can be compared with classical energy KT. (thermal energy $\approx KT$)

Limiting forms for quantum limit:

($h\nu << KT$ is the classical limit)

for $h\nu >> KT$

$$U(\nu)\,d\nu = \frac{8\pi h\nu^3}{c^3}\exp\left(-\frac{h\nu}{KT}\right)d\nu$$

This is known as Wein's formula.
or Wein's displacement law

$$I_\lambda = \frac{2hc^2}{\lambda^5}\frac{1}{\left[\exp\left(\frac{hc}{\lambda kT}\right)-1\right]}$$

$$\frac{dI_\lambda}{d\lambda} = 0$$

Let $x = \frac{hc}{\lambda kT}$, $\lambda = \frac{hc}{xkT} \Rightarrow \lambda^5 = \left(\frac{hc}{xkT}\right)^5 \Rightarrow \lambda^{-5} = \left(\frac{xkT}{hc}\right)^5$

$$I_\lambda \propto \frac{x^5}{e^x-1}$$

$$I_\lambda = 2hc^2 \cdot \lambda^{-5}\frac{1}{(\exp x-1)}$$
$$= 2hc^2 \alpha \left(\frac{xkT}{hc}\right)^5 \frac{1}{(\exp x-1)}$$
$$= 2hc^2 \alpha \frac{x^5k^5T^5}{h^5c^4}\frac{1}{(\exp x-1)}$$

$$= \frac{2x^5k^5T^5}{h^4c^3}\frac{1}{(\exp x-1)}$$
$$= \left(\frac{2k^5T^5}{h^4c^3}\right)\frac{x^5}{(\exp x-1)}$$

$$I_\lambda = \underbrace{\frac{2(kT)^5}{h^4c^3}}_{\text{constant for fixed } T} \cdot \frac{x^5}{(\exp x-1)} \Rightarrow I_\lambda \propto \frac{x^5}{\exp x-1}$$

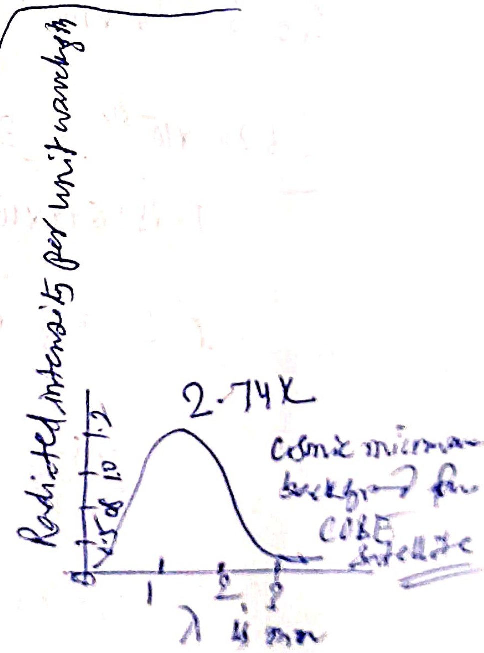
$$f(x) = \frac{x^5}{e^x-1} \Rightarrow \frac{d}{dx}\left(\frac{x^5}{e^x-1}\right) = 0 \Rightarrow 5(e^x-1) = xe^x$$

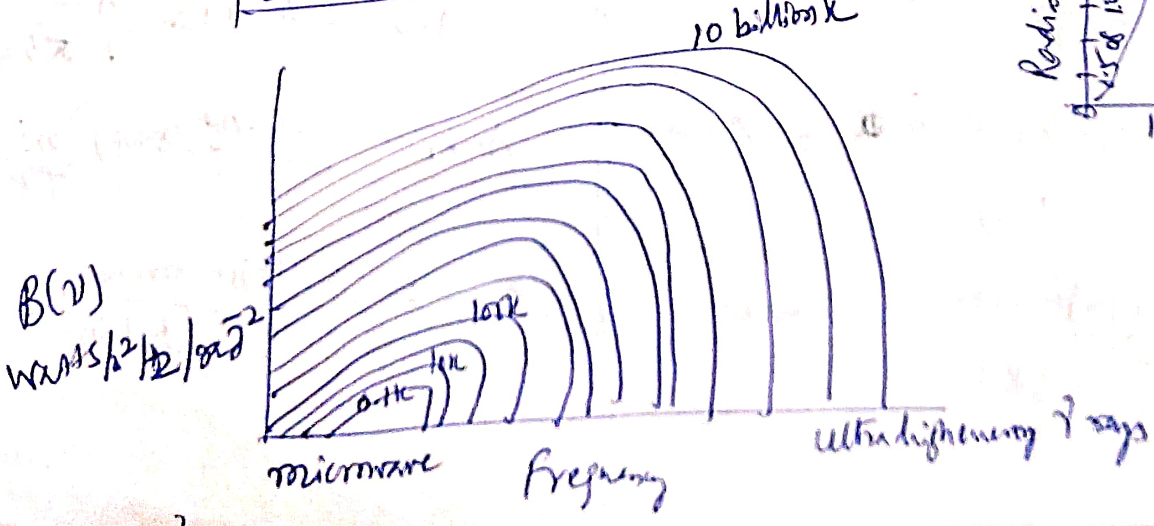
transcedental eqn $x = 4.965$

$$\frac{hc/\lambda_{max}}{kT} = 4.965 \quad \text{(but not one)}$$

$\lambda_{max} T \cong 2.898 \times 10^{-3}$ m K

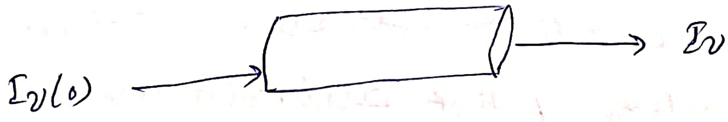
$$\boxed{\lambda_{max} T \approx 3 \text{ mm K}}$$





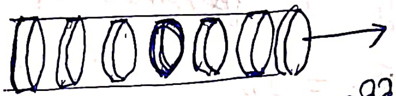
(*) Let us relook at the radiative transfer eqn

$$I_\nu = I_\nu(0)\, e^{-\tau_\nu} + I_\nu^B\,[1 - e^{-\tau_\nu}]$$



1st term : $\boxed{I_\nu = I_\nu(0)\, e^{-\tau_\nu}}$ incident intensity is attenuated by the medium

2nd term $I_\nu = + I_\nu^B [1 - e^{-\tau_\nu}]$



The medium itself add some radiation to the existing radiation (passing one). But part of it can also absorbed by itself. ($e^{-\tau_\nu}$)

$$I_\nu^B - I_\nu^B e^{-\tau_\nu} = I_\nu^B (1 - e^{-\tau_\nu})$$

✓ **Radiation from a layer of material:**

temperature T

If there is no specific intensity falls on this material, what would be the I_ν comes out from this material, which is at temperature K and is at LTE

$I_\nu = I_\nu^B [1 - e^{-\tau_\nu}]$; assume no background emission — 1st term $= 0$

If $\tau_\nu \gg 1$, $\frac{1}{e^{\tau_\nu}} \to 0 \Rightarrow \boxed{I_\nu = I_\nu^B}$

If $\tau_\nu \ll 1$, $e^{-\tau_\nu} = e^{-x} \simeq 1 - x = (1 - \tau_\nu)$

$$I_\nu = I_\nu^B [1 - (1 - \tau_\nu)]$$

$$= I_\nu^B \tau_\nu$$

$$\boxed{I_\nu = I_\nu^B}$$

$\boxed{I_\nu = \tau_\nu I_\nu^B} \Rightarrow$ fraction of I_ν^B would be observed at the

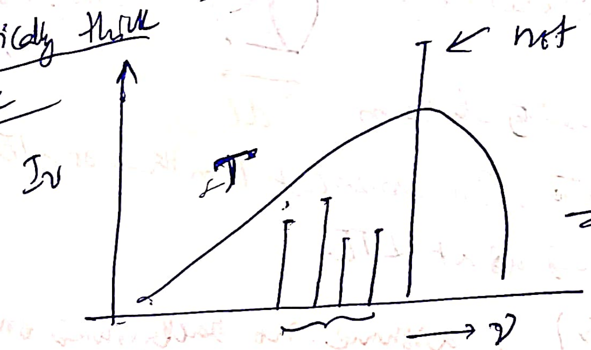
(*) specific intensity that we observe in and optically thin case is always a fraction of τ_ν but less than I_ν^B (which is when $\tau_\nu \gg 1$)

Therefore, for $I_\nu = I_\nu^B[1-e^{-\tau_\nu}]$

$\boxed{I_\nu = I_\nu^B \text{ for } \tau_\nu >> 1}$ and $\boxed{I_\nu = \tau_\nu I_\nu^B}$, $\tau_\nu << 1$

* For optically thin body, the radiation intensity is always less than that from a blackbody at that particular frequency.
* as the τ_ν increases, $\tau_\nu I_\nu^B$ also increases, but reaches to I_ν^B at some optical depth.
* If $\tau_\nu >> 1$, then eventually I_ν becomes equal to blackbody specific intensity $I_\nu \approx I_\nu^B$ (body is opaque)
* we should notice that I_ν would never exceeds I_ν^B ~~for~~ even for exceedingly infinite optical depth values. this is known as the "Blackbody limit"

optically thick case

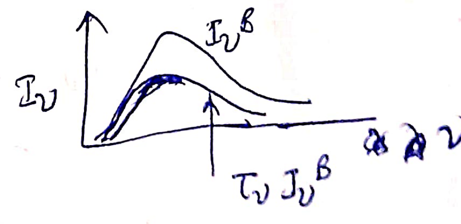


$\Rightarrow I_\nu \approx I_\nu^B$ for $\tau_\nu >> 1$

these levels of I_ν at few ν' values are permissible, but

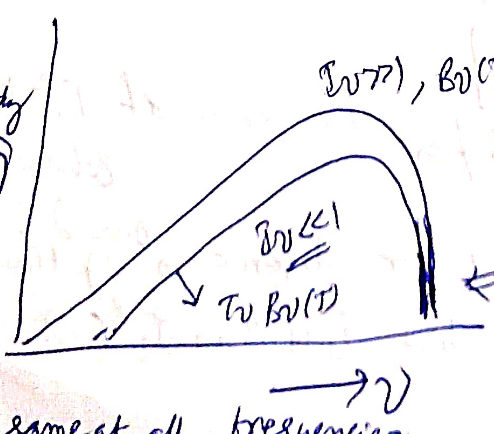
how would a blackbody look like for optically thin case?

$I_\nu = \tau_\nu I_\nu^B$ for $\tau_\nu << 1$



$\tau_\nu >> 1$ true blackbody

$\tau_\nu << 1$: diluted blackbody (same shape, lower amplitude) if τ_ν = same at all frequencies



← they do not intersect with each other.

LASER/MASER

LASER case: I_ν exceeds far higher than I_ν^B at T

LASER keeps at room temperature, but I_ν far higher than I_ν^B (T = 300K). Similarly MASER

LASER: Light Amplification by Stimulated Emission of Radiation

MASER: Microwave Amplification by Stimulated Emission of ra

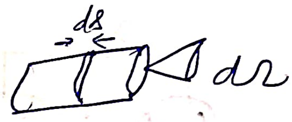
→ generate highly coherent, monochromatic, and directional EM radiation. ← via stimulated emission

Lasers: Visible/UV/IR vs MASER: microwave

Reason: Absorption coefficient α_ν is negative

We know the loss of brightness in a beam as it travels a distance 'ds' is

$$dI_\nu = -\alpha_\nu I_\nu \, ds$$

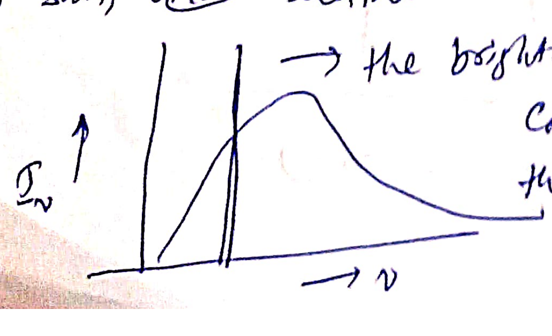


Here, α_ν is positive ⇒ corresponds to the loss of light by convention

In LASER/MASER, the medium amplifies the signal. The absorption coefficient is negative and hence each slab emits the light in a stimulated fashion.

$$dI_\nu = |\alpha_\nu| I_\nu \, ds$$

Laser/Maser violates Blackbody limit because the atoms/molecules involved in it are not in LTE, and the radiation is coherent (while in Sun, radiation from BBs are incoherent)



→ the brightness temperature of a laser could be 10^{20} K, assuming that the emission at that particular frequency is from a hypothetical black body.

$$I_\nu = \underbrace{I_\nu(0)\, e^{-\tau_\nu}}_{\text{absorbed by the medium}} + \underbrace{I_\nu^{S}\left[1 - e^{-\tau_\nu}\right]}_{\text{emission by the medium + self absorption of the emitted emission}}$$

absorbed by the medium

emission by the medium + self absorption ~~by~~ of the emitted emission

Sun at Limb: ① no background emission

$$I_\nu = I_\nu(0)\, e^{-\tau} = 0$$

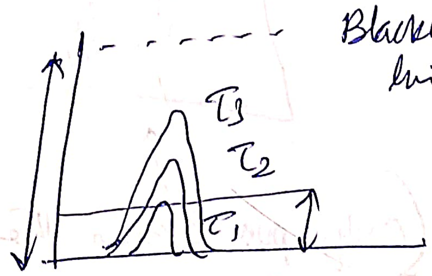
② & no solar atmosphere is optically thin $\tau_\nu \ll 1$

$$1 - e^{-\tau} = 1 - (1 - \tau + \tau^2 - \tau^3 \ldots)$$

$$= \tau$$

$$I_\nu^S (1 - e^{-\tau_\nu}) = \tau_\nu I_\nu^B$$

Limb has $\tau_\nu I_\nu^B$

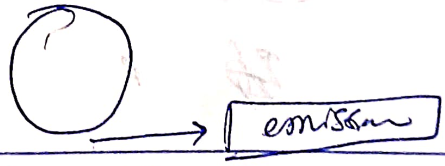


$\tau_1 < \tau_2 < \tau_3$

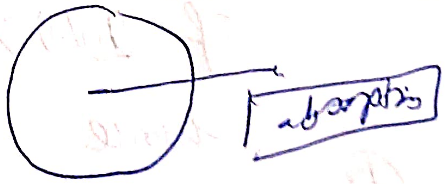
emission

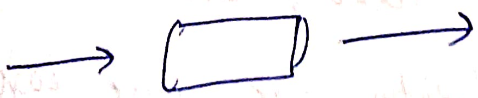
$\tau_1 I_\nu^B < \tau_2 I_\nu^B < \tau_3 I_\nu^B < I_\nu^B$

Chromosphere doesn't ~~acts~~ acts like a black body as there is ~~not~~ ~~as~~ overall much emission.



Central Disk Radiation passes from interior Sun.





$$I_\nu = I_\nu(0)\, e^{-\tau_\nu} + I_\nu^B \left(1 - e^{-\tau_\nu}\right)$$

outer layers of Sun are optically thin $\tau_\nu \ll 1$

I_ν ~~$= I_\nu(0) - \tau_\nu$~~

$e^{-\tau_\nu} = 1 - \tau_\nu$ (for $\tau_\nu << 1$) { only first two terms only left

$$I_\nu = I_\nu(0)\, e^{-\tau} + I_\nu^B \left[1 - e^{-\tau_\nu}\right]$$

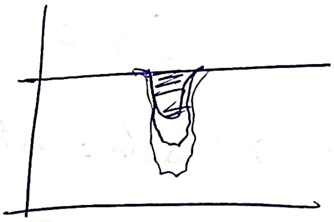
$$= I_\nu(0)\left[1 - \tau_\nu\right] + I_\nu^B\left[1 - (1 - \tau_\nu)\right]$$

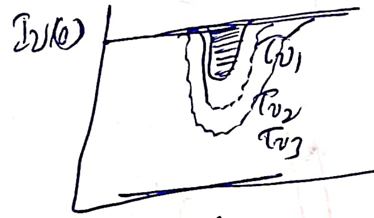
$$= I_\nu(0) - I_\nu(0)\,\tau_\nu + \tau_\nu I_\nu^B$$

$$I_\nu \cong I_\nu(0) - \tau_\nu\left[I_\nu(0) - I_\nu^B\right]$$

↑

from interior of the Sun





↑

(Only possible when the Background emission > foreground emission)

less radiation is taken away ↖ $\tau_{\nu 1} < \tau_{\nu 2} < \tau_{\nu 3}$ → more radiation of background emission is taken

depth is because of the optical depth of the photospheric radiation

$\boxed{I_\nu(0) - \tau_\nu(I_\nu(0))}$ at diff τ_ν values are shown

We get the absorption only when the second component is +ve within the bracket $[I_\nu(0) - I_\nu^B]$

$=$ Background emission $-$ foreground emission

If $I_\nu(0) > I_\nu(B)$, then $I_\nu(0) - I_\nu^B = +ve$,

hence $-\tau_\nu[I_\nu(0) - I_\nu^B]$ is taken away from background to cause absorption feature

Ex: (1) Vehicle Head lights are more bright than the cycle torch $\Rightarrow$ we can only see ~~byc~~ bicycle in absorption / Silhouette

(2) firefly can ~~not~~ only be seen in the night time but not in the day time.